My Project

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Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

contentsline/text/before4 1DoublyLinkedList.h 7DoublyLinkedList.h contentsline/text/after4 1DoublyLinkedList.h 7toc/leaders/beforetoc/leaders toc/leaderstoc/leaders/aftercontentsline/page/before4 17 7contentsline/page/after4 17
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Chapter 3

Class Documentation

3.1 DoublyLinkedList< T > Class Template Reference

Public Member Functions

- bool **isEmpty** () const
- void **addFront** (const T &val)
- void **addBack** (const T &val)
- void **removeFront** ()
- void **removeBack** ()
- void **clear** ()
- void **display** () const
- ListIterator< T > **getIterator** () const
- void **displayReverse** () const

The documentation for this class was generated from the following file:

- DoublyLinkedList.h

3.2 ListFactory< T > Class Template Reference

Static Public Member Functions

- static DoublyLinkedList< T > * **createList** ()

The documentation for this class was generated from the following file:

- ListFactory.h

3.3 ListIterator< T > Class Template Reference

Public Member Functions

- **ListIterator** (Node< T > *startNode)
- bool **hasNext** () const
- bool **hasPrev** () const
- T **next** ()
- T **prev** ()
- T **getCurrent** () const
- bool **isValid** () const

The documentation for this class was generated from the following file:

- Iterator.h

3.4 Node< T > Struct Template Reference

Public Member Functions

- **Node** (const T &wartosc)

Public Attributes

- T **data**
- Node * **next**
- Node * **prev**

The documentation for this struct was generated from the following file:

- Node.h

Chapter 4

File Documentation

4.1 DoublyLinkedList.h

```
00001 #ifndef DOUBLY_LINKED_LIST_H
00002 #define DOUBLY_LINKED_LIST_H
00003
00004 #include <iostream>
00005 #include "Node.h"
00006 #include "Iterator.h"
00007
00008 // Główna klasa listy dwukierunkowej działająca na stercie
00009 template <typename T>
00010 class DoublyLinkedList {
00011 private:
00012     Node<T>* head; // początek listy
00013     Node<T>* tail; // koniec listy
00014     int count;     // licznik elementów
00015
00016 public:
00017     // Konstruktor - tworzy pustą listę
00018     DoublyLinkedList() {
00019         head = nullptr;
00020         tail = nullptr;
00021         count = 0;
00022     }
00023
00024     // Destruktor - czyszczenie pamięci
00025     ~DoublyLinkedList() {
00026         clear();
00027     }
00028
00029     // Sprawdzenie czy lista jest pusta
00030     bool isEmpty() const {
00031         return head == nullptr;
00032     }
00033
00034     // Dodanie elementu na początek listy
00035     void addFront(const T& val) {
00036         Node<T>* newNode = new Node<T>(val);
00037         if (isEmpty()) {
00038             head = tail = newNode;
00039         } else {
00040             newNode->next = head;
00041             head->prev = newNode;
00042             head = newNode;
00043         }
00044         count++;
00045     }
00046
00047     // Dodanie elementu na koniec listy
00048     void addBack(const T& val) {
00049         Node<T>* newNode = new Node<T>(val);
00050         if (isEmpty()) {
00051             head = tail = newNode;
00052         } else {
00053             tail->next = newNode;
00054             newNode->prev = tail;
00055             tail = newNode;
00056         }
00057         count++;
00058     }
```

```
00059
00060     // Usunięcie elementu z początku listy
00061     void removeFront() {
00062         if (isEmpty()) return;
00063         Node<T>* temp = head;
00064         head = head->next;
00065         if (head != nullptr) {
00066             head->prev = nullptr;
00067         } else {
00068             tail = nullptr; // lista stała się pusta
00069         }
00070         delete temp;
00071         count--;
00072     }
00073
00074     // Usunięcie elementu z końca listy
00075     void removeBack() {
00076         if (isEmpty()) return;
00077         Node<T>* temp = tail;
00078         tail = tail->prev;
00079         if (tail != nullptr) {
00080             tail->next = nullptr;
00081         } else {
00082             head = nullptr; // lista stała się pusta
00083         }
00084         delete temp;
00085         count--;
00086     }
00087
00088     // Czyszczenie całej listy
00089     void clear() {
00090         while (!isEmpty()) {
00091             removeFront();
00092         }
00093     }
00094
00095     // Wyświetlanie listy od początku
00096     void display() const {
00097         Node<T>* current = head;
00098         std::cout « "Lista: ";
00099         while (current != nullptr) {
00100             std::cout « current->data « " ";
00101             current = current->next;
00102         }
00103         std::cout « std::endl;
00104     }
00105     // Zwraca iterator początkowy listy
00106     ListIterator<T> getIterator() const {
00107         return ListIterator<T>(head);
00108     }
00109     // Wyświetlanie listy od końca (w odwrotnej kolejności)
00110     void displayReverse() const {
00111         Node<T>* current = tail;
00112         std::cout « "Lista od tylu: ";
00113         while (current != nullptr) {
00114             std::cout « current->data « " ";
00115             current = current->prev;
00116         }
00117         std::cout « std::endl;
00118     }
00119 };
00120
00121 #endif
```

4.2 Iterator.h

```
00001 #ifndef ITERATOR_H
00002 #define ITERATOR_H
00003
00004 #include "Node.h"
00005
00006 // Wzorzec Iteratora do przechodzenia po liście dwukierunkowej
00007 template <typename T>
00008 class ListIterator {
00009 private:
00010     Node<T>* current; // aktualny węzeł, na którym stoi iterator
00011
00012 public:
00013     // Konstruktor przyjmujący początkowy węzeł
00014     ListIterator(Node<T>* startNode) {
00015         current = startNode;
00016     }
00017
```

```
00018     // Sprawdzenie czy jest następny element
00019     bool hasNext() const {
00020         return current != nullptr && current->next != nullptr;
00021     }
00022
00023     // Sprawdzenie czy jest poprzedni element
00024     bool hasPrev() const {
00025         return current != nullptr && current->prev != nullptr;
00026     }
00027
00028     // Przejście do następnego elementu i zwrócenie jego wartości
00029     T next() {
00030         if (current != nullptr && current->next != nullptr) {
00031             current = current->next;
00032         }
00033         return current ? current->data : T();
00034     }
00035
00036     // Przejście do poprzedniego elementu i zwrócenie jego wartości
00037     T prev() {
00038         if (current != nullptr && current->prev != nullptr) {
00039             current = current->prev;
00040         }
00041         return current ? current->data : T();
00042     }
00043
00044     // Pobranie wartości aktualnego elementu
00045     T getCurrent() const {
00046         return current ? current->data : T();
00047     }
00048
00049     // Sprawdzenie czy iterator wskazuje na poprawny element
00050     bool isValid() const {
00051         return current != nullptr;
00052     }
00053 };
00054
00055 #endif
```

4.3 ListFactory.h

```
00001 #ifndef LIST_FACTORY_H
00002 #define LIST_FACTORY_H
00003
00004 #include "DoublyLinkedList.h"
00005
00006 // Wzorzec Projektowy Factory (Fabryka) do tworzenia listy na stercie
00007 template <typename T>
00008 class ListFactory {
00009 public:
00010     static DoublyLinkedList<T>* createList() {
00011         return new DoublyLinkedList<T>();
00012     }
00013 };
00014
00015 #endif
```

4.4 Node.h

```
00001 #ifndef NODE_H
00002 #define NODE_H
00003
00004 // Pojedynczy element listy dwukierunkowej
00005 template <typename T>
00006 struct Node {
00007     T data;         // wartość
00008     Node* next;     // wskaźnik na następny element
00009     Node* prev;     // wskaźnik na poprzedni element
00010
00011     // Konstruktor do tworzenia węzła
00012     Node(const T& wartosc) {
00013         data = wartosc;
00014         next = nullptr;
00015         prev = nullptr;
00016     }
00017 };
00018
00019 #endif
```